

Veet Zaveri

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Education

Johns Hopkins University

Master of Science in Computer Science

Cumulative GPA: 4.00

Baltimore, MD

Expected December 2027

DePaul University

Bachelor of Science in Business Management with Honors

Cumulative GPA: 3.99/4.00

Chicago, IL

June 2021

Research Focus

Reliable and generalizable machine learning, with emphasis on adversarial robustness, evaluation under distribution shift, multimodal learning, foundation-model evaluation, and representation learning.

Publications

Towards a Better Characterization of Adversarial Attacks in Geospatial Imagery

Remote Sensing, 2026

First author. Evaluated 20+ vision architectures under adversarial attacks, domain shift, unseen attack variants, and low-prevalence screening. DOI: 10.3390/rs18122041

Non-invasive Intracranial Pressure Waveform Reconstruction with Deep Learning

medRxiv, 2026

Co-first author. Under review at *npj Digital Medicine*; externally validated deep learning models for continuous noninvasive ICP waveform reconstruction. DOI: 10.64898/2026.06.07.26354958

A Motion Planning Algorithm in a Figure Eight Track

PUMP Journal, 2023

Equal contribution. Developed a topological motion-planning algorithm for collision-free coordination of two robots. DOI: 10.46787/pump.v6i0.3559

Research & Engineering Experience

Institute for Defense Analyses

Research Associate (CS Researcher / Data Scientist)

Alexandria, VA

Jan. 2024 – Present

- Trained and systematically evaluated **20+ CNN and Transformer architectures** for detecting adversarial and generative manipulation of satellite imagery, analyzing generalization under domain shift, unseen attack variants, and low-prevalence screening; **published the findings as first author** in *Remote Sensing*.
- Developed a machine-learning-powered, parallelized Python acquisition and cleaning pipeline that processed **21M+ records**, increased ingestion throughput by **50×**, and supported **50+ analytical products for senior U.S. government sponsors**; selected as the **sole recipient of IDA's highest-tier 3-Star Analysis Award** for the six-month award cycle.
- Built and deployed an automated data pipeline and **Streamlit application** covering **26K+ records** across multiple sources, with automated weekly refreshes and adoption by sponsors and **20+ internal users**.
- Evaluated and integrated frontier LLMs via **AWS Bedrock** across data-processing workflows; benchmarked performance parameter extraction on **30+ documents**, using few-shot and structured prompt engineering to achieve **90%+ extraction accuracy** while reducing the **hallucination rate to <5%**, and applied LLMs to clean and classify **10K+ address records**.
- Built a semantic document-analysis system over **500+ documents** using sentence-transformer embeddings and cosine similarity to identify semantically overlapping content, reducing manual analysis by **50+ hours**.
- Built an NLP application that classifies **300+ article titles daily** and generates summaries using ktrain and txtai, reducing manual literature-monitoring workloads by **20+ hours per week**.
- Clustered and linked **2,000+ entity names** across heterogeneous datasets using TF-IDF n-grams, sentence-transformer embeddings, and topic modeling for cross-source entity resolution and downstream analysis.
- Engineered and optimized **10+ TB of SQL data infrastructure** on Linux using indexing, partitioning, and database tuning to eliminate API bottlenecks; increased data retrieval throughput by **20×**, from approximately 10K to 200K records per day.

LCICM – Johns Hopkins University

Graduate Deep Learning Researcher

Remote

Aug. 2024 – Present

- Co-first-authored** a study of noninvasive intracranial-pressure waveform reconstruction across **158 adult patients** and approximately **5,322 hours of waveform data**, sharing responsibility for methodology, software, analysis, data curation, visualization, and manuscript preparation; manuscript currently under review at *npj Digital Medicine*.
- Developed and evaluated **LSTM, GRU, multiscale V-Net, TCN, and Transformer-based models** for continuous ICP reconstruction from ABP, PPG, and ECG, including early- and late-fusion strategies and waveform-aware training objectives.
- Conducted independent cross-institutional validation from a **47-patient internal cohort** to a **111-patient external cohort**; performance across five architectures ranged from **4.276–4.946 mmHg MAE** and **0.599–0.722 Pearson correlation**.
- Built an end-to-end physiological signal-processing pipeline to synchronize concurrent waveforms, construct **30-second prediction windows**, and resample high-frequency signals to **125 Hz** for model-ready tensors.
- Developed automated signal-quality filtering for flatline, noisy, periodic, and physiologically implausible waveform segments using amplitude, variability, heart-rate plausibility, and signal-quality checks.

LCICM – Johns Hopkins University
Lab Director

Remote
Sep. 2026 – Present

- Help supervise **5 active research projects involving 11 students**, providing guidance on research planning, technical challenges, project execution, and next-step prioritization.
- Coordinate weekly laboratory meetings and project reviews for an interdisciplinary computational critical-care research group, organizing student updates around completed work, current problems, and next steps.
- Support research administration, IRB-related coordination, and broader laboratory operations under Principal Investigator Robert D. Stevens.

Wilbur Wright College
Topology Student Researcher

Chicago, IL
Jan. 2022 – Jun. 2023

- Developed a topological approach to collision-free motion planning for two robots on a figure-eight track, proving the problem has **topological complexity 3** and constructing an explicit algorithm with **three continuous motion-planning instructions**.
- **Co-first-authored** “A Motion Planning Algorithm in a Figure Eight Track,” published in *The PUMP Journal of Undergraduate Research* (2023), applying configuration-space topology to characterize instabilities in robotic motion-planning algorithms.

Additional Engineering Experience

Mayo Clinic
IT Software Engineer Intern

Remote
Jul. 2023 – Dec. 2023

- Contributed to the development and internal deployment of a **specimen-tracking application** at Mayo Clinic using **TypeScript, React, HTML, and CSS** as part of an Agile Scrum engineering team.
- Evaluated modern front-end frameworks and build tools, including **Next.js** and **Vite**, and developed technical comparisons to guide architecture and technology selection for future software projects.
- Integrated the **Microsoft Graph API** into the specimen-tracking application to programmatically retrieve and display authenticated user profile information and photos.

Astoria.ai
Software Engineer Intern / Software Architecture Team Lead

Remote
Sep. 2022 – May 2023

- Led the Software Architecture team in evaluating **Redis, Kafka, Fluvio, and Hadoop** for a talent-management platform aggregating, correlating, and transforming **10M+ records**.
- Built RESTful ingestion and web-scraping services that collected and processed **100K+ job postings** for downstream search, analytics, and candidate-matching workflows.
- Applied regression, clustering, predictive modeling, and neural networks to candidate-matching pipelines and developed full-stack features in Python, Java, JavaScript, HTML, and CSS.

Teaching Experience

Johns Hopkins University
Data Structures TA / Grader

Baltimore, MD
Jan. 2023 – May 2023

- Evaluated Java and Python coursework for **20+ students** and provided individualized feedback on **data structures, algorithms, debugging, and software-development practices**.

Selected Projects

Braille Display Reader – HopHacks Hackathon

Sep. 2022

- Placed **top 5 out of 32 teams** at HopHacks 2022 by developing an Arduino-based Braille display capable of rendering characters from arbitrary text input.
- Integrated an **MPU6050 accelerometer** with Python-based motion processing to detect gestures and dynamically navigate characters displayed on the LED matrix.

Technical Skills

Programming: Python, Java, TypeScript/JavaScript, SQL **ML:** PyTorch, TensorFlow, Keras, scikit-learn, Sentence Transformers
Research: Deep learning, LLM evaluation, computer vision, multimodal learning, time-series modeling, adversarial robustness, distribution-shift analysis, representation learning

Data & Systems: Pandas, NumPy, SciPy, PostgreSQL, SQL Server, MongoDB, AWS, Git, Docker, Kafka, Redis, Linux